

DEPARTMENT OF CSE
CS T61 ENTERPRISE SOLUTIONS
Unit-III – ORACLE SUITE
QUESTION BANK WITH ANSWER

Oracle Suite : Oracle Apps 11i - Application Framework - File System - Workflow Analysis - SQL / PLSQL fundamentals - Creating Forms - Oracle Reports. Oracle Electronic Data Interchange – functions of EDI – Data File Structure - Oracle Data, Oracle Database - Oracle Database - DW vs OLTP - DW Connectors.

PART A (2 MARKS)

1. What is Oracle Application Architecture?

The Oracle Applications Architecture is a framework for multi-tiered, distributed computing that supports Oracle Applications products. In this model, various servers are distributed among multiple levels, or tiers.

2. List the three tiers in an Oracle E-Business Suite installation.

- **Database tier**, which supports and manages the Oracle database
- **Application tier**, which supports and manages the various Applications components, and is sometimes known as the middle tier
- **Desktop tier**, which provides the user interface via an add-on component to a standard web browser

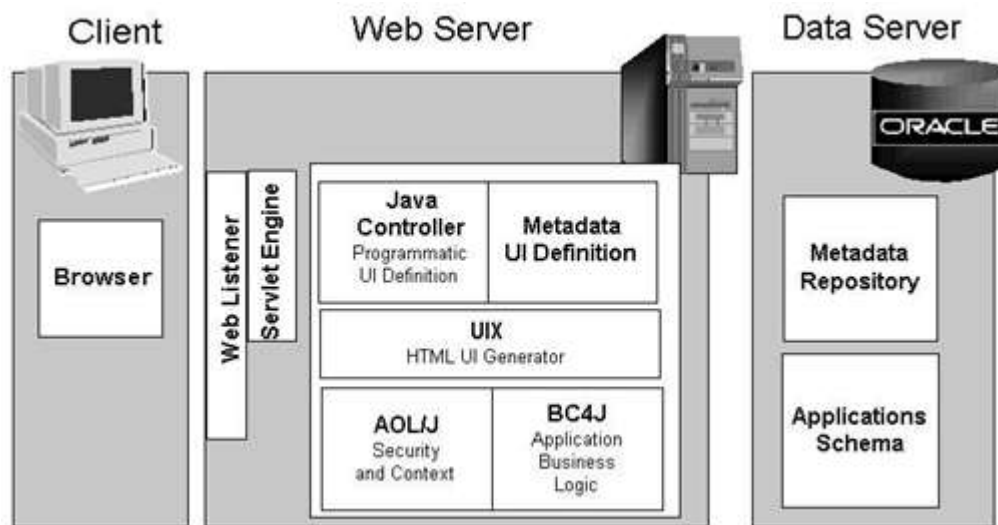
3. Define Web server.

- The Oracle HTTP server (powered by Apache) acts as the Web server.
- It processes the requests received over the network from the desktop clients, and includes additional components such as:
 - Web Listener
 - Java Servlet Engine
 - Java Server Pages (JSP)

4. What is Oracle Application Framework?

- The Oracle Applications Framework is the development platform for HTML-based applications.
- It consists of a Java-based application tier framework and associated services, designed to facilitate the rapid deployment of HTML-based applications.

5. Draw Oracle Applications Framework Architecture.



6. Write a note on Discoverer Server.

Discoverer is an ad hoc query, reporting, analysis, and publishing tool that allows business users at all levels of an organization to gain immediate access to information from data marts, data warehouses, and online transaction processing (OLTP) systems.

7. What are operations from Admin server?

- Upgrading Oracle Applications
- Applying database patches to Oracle Applications
- Maintaining Oracle Applications data

8. Write a note on Daily Business Intelligence (DBI).

- Daily Business Intelligence (DBI) is a reporting framework that is integrated with Oracle E-Business Suite.
- It replaces the Business Intelligence System (BIS), and includes a new set of materialized views that pre-summarize transaction data. Using Daily Business Intelligence overview pages, managers can view summarized information across multiple organizations, drilling down to specific transaction details on a daily basis.

9. Define oracle overflow.

Oracle Workflow provides an infrastructure for the enterprise-wide communication of data related to defined business events, providing the capabilities needed to:

- Manage enterprise business processes that may span trading partners
- Support standard and personalized business rules
- Streamline and automate transaction flows
- Manage exceptions without manual intervention

10. What is Oracle Workflow Business Event System and oracle workflow builder?

Oracle Workflow Business Event System

- It provides a workflow-enabled solution for your enterprise application integration requirements.
- The Business Event System supports the following types of integration:
 - Message-based point-to-point system integration
 - System integration messaging hubs
 - Distributed applications messaging

Oracle workflow builder

- Oracle Workflow Builder provides a graphical drag and drop process designer.
- We can create and evolve business processes to incorporate existing business practices between your organization and customers or suppliers, without modifying existing business processes and without changing applications code.

11. Define Oracle Workflow Event Manager.

- It enables registration of significant business events for selected applications, including functions that generate the XML event messages.

12. List the features in Oracle E-Business Suite Release 11i.

Release 11i utilizes various Oracle database features. They are

- Optimize performance
- Scalability, and
- Business intelligence capacity

13. Mention the use of cost-based optimization (CBO).

- The Oracle optimizer evaluates many factors to calculate the most efficient way to execute a SQL statement.
- The optimizer compares the costs of the execution plans and chooses the one with the smallest cost, i.e. optimum execution characteristics.

14. Draft a note on Real Application Clusters.

- Real Application Clusters (RAC) harness the processing power of multiple interconnected computers.
- RAC software and a collection of computers (known as a cluster) harness the processing power of each component to create a robust and powerful computing environment.
- A large task divided into subtasks and distributed among multiple nodes is completed more quickly and efficiently than if the entire task was processed on one node.
- Cluster processing also facilitates deployment of additional hardware resources for larger workloads and rapidly growing user populations.

15. Define Structured Query Language (SQL).

- An internationally standardized language that is used to access data in a relational database.
- SQL statements enable you to perform the following tasks:
 - Query data
 - Insert, update, and delete rows in a table
 - Create, replace, alter, and drop objects
 - Control access to the database and its objects
 - Guarantee database consistency and integrity

16. Write short notes on Data Definition Language (DDL).

Data definition language (DDL) statements define, structurally change, and drop schema objects. For example, DDL statements enable you to:

- Create, alter, and drop schema objects and other database structures, including the database itself and database users. Most DDL statements start with the keywords CREATE, ALTER, or DROP.
- Delete all the data in schema objects without removing the structure of these objects (TRUNCATE)
- Grant and revoke privileges and roles (GRANT, REVOKE).
- Turn auditing options on and off (AUDIT, NOAUDIT).
- Add a comment to the data dictionary (COMMENT).

17. Define Data Manipulation Language (DML).

DML statements are the most frequently used SQL statements and enable you to:

- Retrieve or fetch data from one or more tables or views (SELECT).
- Add new rows of data into a table or view (INSERT) by specifying a list of column
- Values or using a subquery to select and manipulate existing data.
- Change column values in existing rows of a table or view (UPDATE).
- Update or insert rows conditionally into a table or view (MERGE).
- Remove rows from tables or views (DELETE)

18. What is PL/SQL?

- PL/SQL provides a server-side, stored procedural language that is easy-to-use, seamless with SQL, robust, portable, and secure.
- We can access and manipulate database data using procedural schema objects called PL/SQL program units.

19. List the advantages of PL/SQL subprograms.

- Improved performance
- Memory allocation
- Improved productivity
- Security with definer's rights procedures

20. What is PL/SQL Packages?

- A PL/SQL package is a group of related subprograms, along with the cursors and use, stored together in the database for continued use as a unit.

21. List out the Advantages of PL/SQL Packages.

- Encapsulation
- Data security
- Better Performance

22. Define Cursor.

- A cursor is a pointer to this context area. PL/SQL controls the context area through a cursor. A cursor holds the rows (one or more) returned by a SQL statement.
- The set of rows the cursor holds is referred to as the active set.

23. What is Electronic Data Interchange (EDI)?

- EDI is an electronic exchange of information between trading partners.
- Data files are exchanged in a standard format to minimize manual effort, speed data processing, and ensure accuracy.

24. List out the functions that can be performed by EDI.

- Define trading partner groups and trading partner locations.
- Enable transactions for trading partners.
- Provide location code conversion between trading partner location codes and codes used in Oracle Applications.
- Provide general code conversion between trading partner codes or standard codes.
- Define interface data files so that application data can interface with EDI translators.

25. Write a note on Data File Structure.

- Oracle Database creates a data file for a table space by allocating the specified amount of disk space plus the overhead for the data file header.
- The operating system under which Oracle Database runs is responsible for clearing old information and authorizations from a file before allocating it to the database

- The data file header contains metadata about the data file such as its size and checkpoint SCN. Each header contains an absolute file number and a relative file number.
- The absolute file number uniquely identifies the data file within the database. The relative file number uniquely identifies a data file within a table space.

26. List some unstructured data.

- Spatial data
- Multimedia data
- Text data

27. What is meant by LOB?

- The large object (LOB) data types enable you to store and manipulate large blocks of unstructured data in binary or character format.
- LOBs provide efficient, random, piece-wise access to the data.

28. Write a short note on external LOBs.

- A BFILE (binary file LOB) is an external LOB because the database stores a pointer to a file in the operating system. The external data is read-only.
- Overview of XML in Oracle Database
- Oracle XML DB is a set of Oracle Database technologies related to high-performance XML storage and retrieval. XML DB provides native XML support by encompassing both SQL and XML data models in an interoperable manner.

29. What is Oracle Spatial and Oracle multimedia?

Oracle Spatial

- Oracle Spatial makes spatial data management easier to users of location-enabled applications and geographic information system (GIS) applications.

Oracle Multimedia

- Oracle Multimedia enables Oracle Database to store, manage, and retrieve images, medical images that follow the Digital Imaging and Communications in Medicine (DICOM) standard, audio, and video data in an integrated fashion with other enterprise information.

30. Mention the advantages of Oracle Text.

- Oracle Text allows text searches to be combined with regular database searches in a single SQL statement. The Text index is in the database, and Text queries are run in the Oracle Database process.
- The optimizer can choose the best execution plan for any query, giving the best performance for ad hoc queries involving Text and structured criteria.

31. Define DBMS and RDBMS.

DBMS

- A database management system (DBMS) is software that controls the storage, organization, and retrieval of data

RDBMS

- An RDBMS moves data into a database, stores the data, and retrieves it so that it can be manipulated by applications.
- Oracle Database is an RDBMS. RDBMS that implements object-oriented features such as user-defined types, inheritance, and polymorphism is called an object-relational database management system (ORDBMS).

32. What is transaction management?

- Oracle Database is designed as a multiuser database.
- The database must ensure that multiple users can work concurrently without corrupting one another's data.

33. What is meant by Data warehousing?

- A data warehouse is a relational database designed for query and analysis rather than for transaction processing.
- For example, a data warehouse could track historical stock prices or income tax records.

- A warehouse usually contains data derived from historical transaction data, but it can include data from other sources.

34. List the Characteristics of Data warehouse?

- Subject-Oriented
- Nonvolatile
- Time-Variant

35. Write Data warehouse Vs OLTP?

Characteristics	Data warehouse	OLTP
Workload	Designed to accommodate adhoc queries. We may not know the workload of our data warehouse in advance, so it should be optimized to perform well for a wide variety of possible queries	Supports only predefined operations. Our applications might be specifically tuned or designed to support only these operations.
Data Modifications	Updated on a regular basis by the ETL process using bulk data modification techniques. End user of a data warehouse do not directly update the database	Subjected to individual DML statements routinely issued by end users. The OLTP database is always up to date and reflects the current state of each business transaction.
Scheme design	Uses denormalized or partially denormalized schemes (such as star scheme) to optimize query performance.	Uses fully normalized schemes to optimize DML performance and to guarantee data consistency

PART – B (11 MARKS)

1. Explain in detail about Oracle Apps 11i and Oracle Application Framework.

Oracle Suite: Oracle Apps 11i

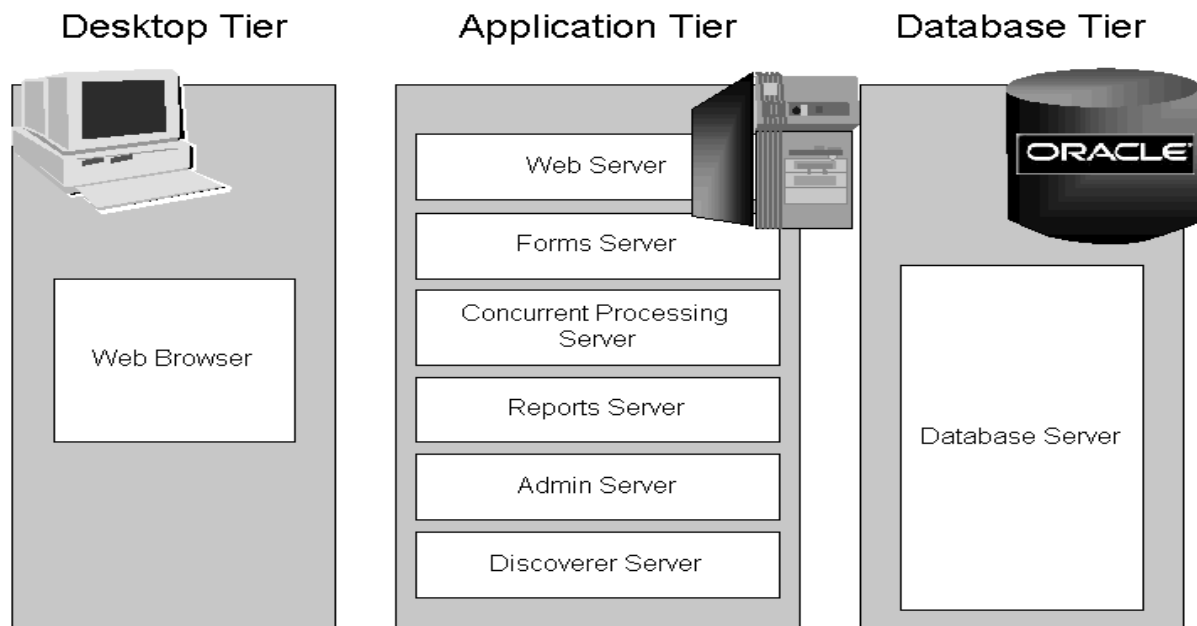
- Oracle Applications comprise the applications software or business software of the Oracle Corporation.
- Oracle's E-Business Suite (also known as Applications/Apps or EB-Suite/EBS) consists of a collection of enterprise resource planning (ERP), customer relationship management (CRM), and supply-chain management (SCM) computer applications either developed or acquired by Oracle. Enterprise applications available as part of Oracle e-Business Suite.



- New 11i end-to-end Implementations or module add-ons in existing implementations or up gradations to new versions.
- Project management
- Requirement Analysis
- Solution Function design
- Gap Fitment Analysis
- Customization Development
- Comprehensive User Training
- Post implementation Support & Maintenance

ORACLE APPLICATIONS ARCHITECTURE

- Oracle Applications 11i eBusiness Suite (Oracle ERP) use the n-tier architecture (the architecture made for the web).
- This architecture can be seen on the following graphic.



Oracle Applications Architecture

The Desktop Tier:

- In Oracle Applications Release 11i, each user logs in to Oracle Applications through the E-Business Suite Home Page on a desktop client web browser.
- The E-Business Suite Home Page provides a single point of access to HTML-based applications, Forms-based applications, and Business Intelligence applications. Once logged in via the E-Business Suite Home Page, you need not sign on again to access other parts of the system.
- Oracle Applications does not prompt again for user name and password, even when you navigate to other tools and products

Forms Client Applet:

- The Forms client applet is a general-purpose presentation applet that supports all Oracle Applications Forms-based products, including those with customizations and extensions

Oracle JInitiator:

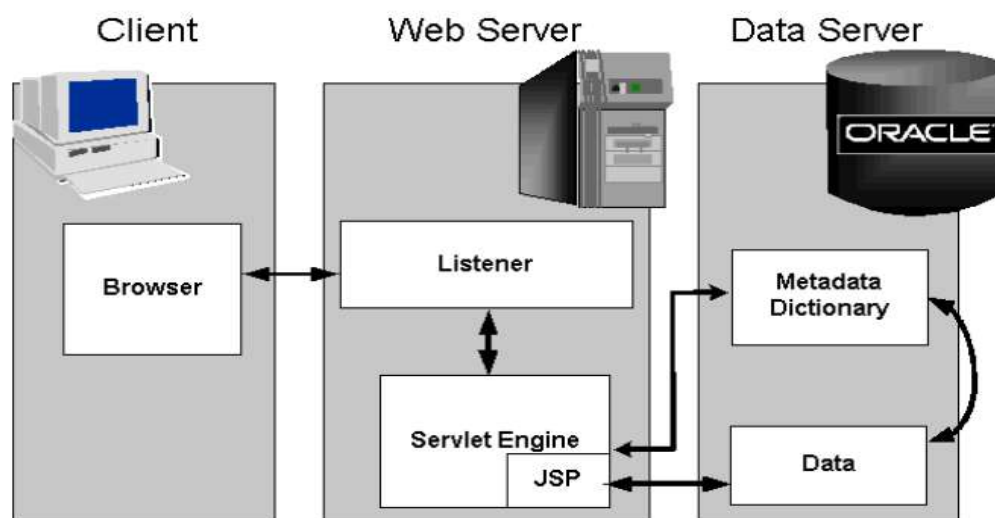
- The Forms client applet must run within a Java Virtual Machine (JVM) on the desktop client. The Oracle JInitiator component allows use of the Oracle JVM on web clients, instead of the browser's own JVM. JInitiator is implemented as a plug-in (Netscape) or ActiveX component (Microsoft Internet Explorer).

Application Tier

- The application tier has a dual role: hosting the various servers that process the business logic, and managing communication between the desktop tier and the database tier.
- This tier is sometimes referred to as the middle tier.
- Oracle9i Application Server comprise the application tier for Oracle Applications
 - Web server
 - Forms server
 - Reports server
 - Discoverer server (optional)
 - Concurrent Processing server
 - Admin server

Web server

- The Oracle HTTP server (powered by Apache) acts as the Web server. It processes the requests received over the network from the desktop clients.



Web server

1. The user clicks the hyperlink of a function from a browser.
2. The browser makes a URL request to the Web listener.
3. The Web listener contacts the Servlet engine (JServ) where it runs a JSP.
4. The JSP obtains the content from the Applications tables and uses information from the metadata dictionary to construct the HTML page.
5. The resulting HTML page is passed back to the browser, via the Web server.

Forms server

- The Forms server hosts the Oracle Applications forms and associated runtime engine that support the professional interface and mediates the communication between the desktop client and the Oracle database server,
- The Forms server communicates with the desktop client using these protocols:
 - Standard HTTP network connection
 - Secure HTTPS network connection
 - TCP/IP connection

Reports server:

- The Reports server dynamically selects the report language at runtime, so users see their reports in their preferred language.

Discoverer Server:

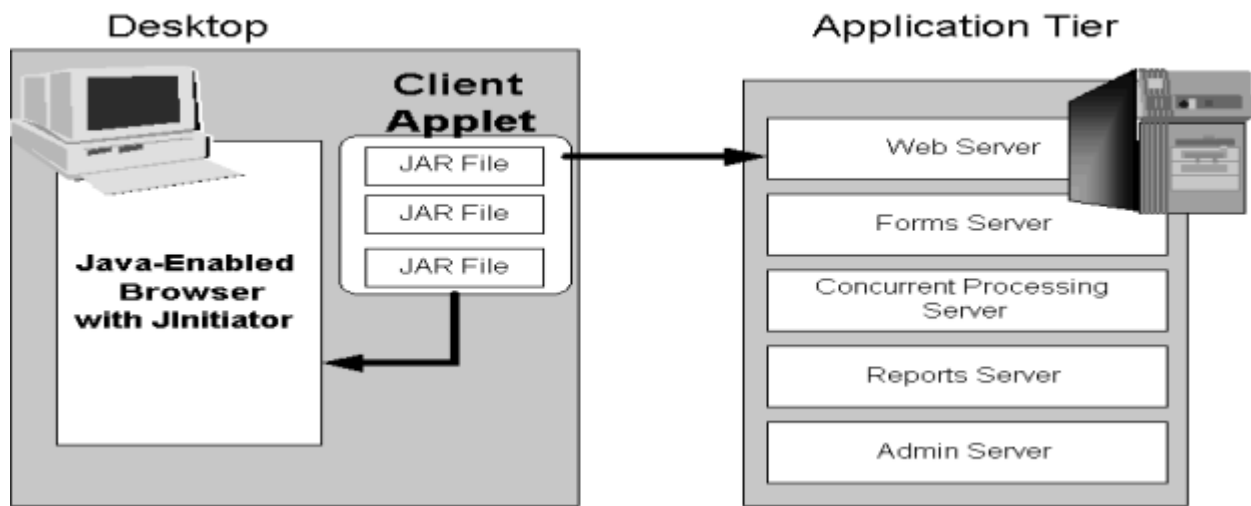
- Discoverer is an ad hoc query, reporting, analysis, and publishing tool that allows business users at all levels of an organization to gain immediate access to information from data marts, data warehouses, and online transaction processing (OLTP) systems.

Admin Server:

- The Admin (Administration) server is located on the node where you maintain the data model and data in your Oracle Applications database.

Database Tier

- The database tier contains the Oracle database server, which stores all the data maintained by Oracle Applications.
- The database also stores the Oracle Applications online help information.
- The database server does not communicate directly with the desktop clients, but rather with the servers on the application tier, which mediate the communications between the database server and the clients.



Database Tier

Oracle Applications DBA (AD):

- The Applications DBA product provides a set of tools for administration of the Oracle Applications file system and database.
- AD tools are used for installing, upgrading, maintaining, and patching the Oracle Applications system.

Oracle Common Modules (AK):

- The Oracle Common Modules can be used to develop inquiry applications for the HTML-based Applications, without the need for any programming.

Oracle Applications Utilities (AU):

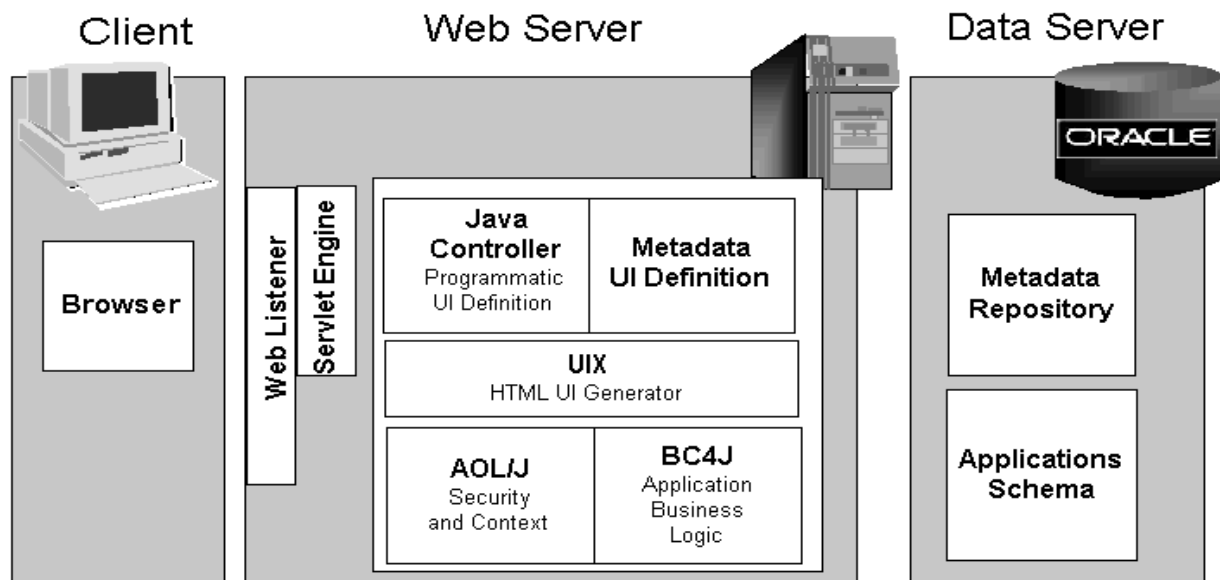
- The Applications Utilities (AU) component is used to maintain the Oracle Applications system. AU hosts a collection of files copied from other products.

ORACLE APPLICATIONS FRAMEWORK ARCHITECTURE

Following is a more detailed description of the processing that occurs in Step 4 of the previous section.

Oracle Applications. Framework uses additional components, including:

- BC4J (Business Components for Java), included in Oracle JDeveloper, is used to create Java business components for representing business logic. It also provides a mechanism for mapping relational tables to Java objects. It allows separation of the application business logic from the UI.
- AOL/J provides the Oracle Applications Framework with underlying security and applications Java services. It provides the Oracle Applications Framework with its connection to the database and with application-specific functionality, such as attachments.



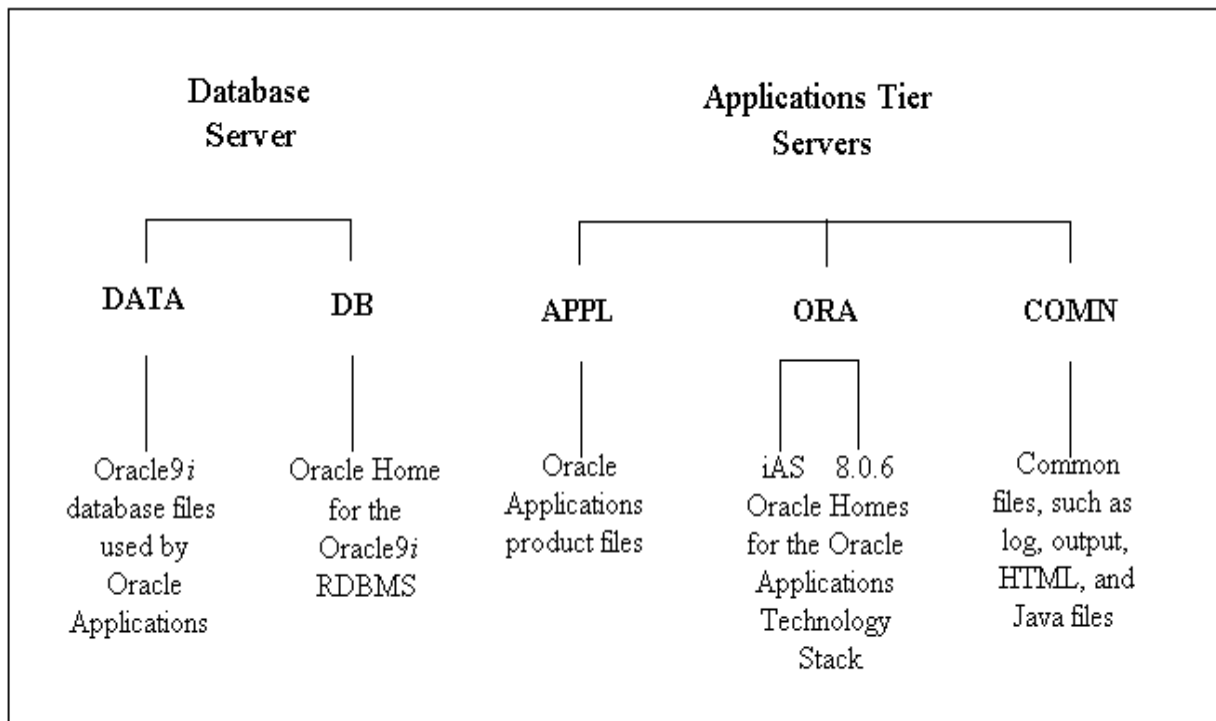
Oracle Applications Framework Architecture

- AOL/J validates user access to the page.
- The page definition is loaded from Metadata Repository on the database tier into the applications tier (Metadata UI Definition).
- The BC4J objects that contain the applications logic and access the database are
- Instantiated.
- The Java Controller programmatically manipulates the page definition as necessary based on dynamic UI rules.
- UIX (HTML UI Generator) interprets the page definition and creates corresponding HTML in accordance with UI standards and sends the page to the browser.

2. Discuss in detail about Oracle File Systems.

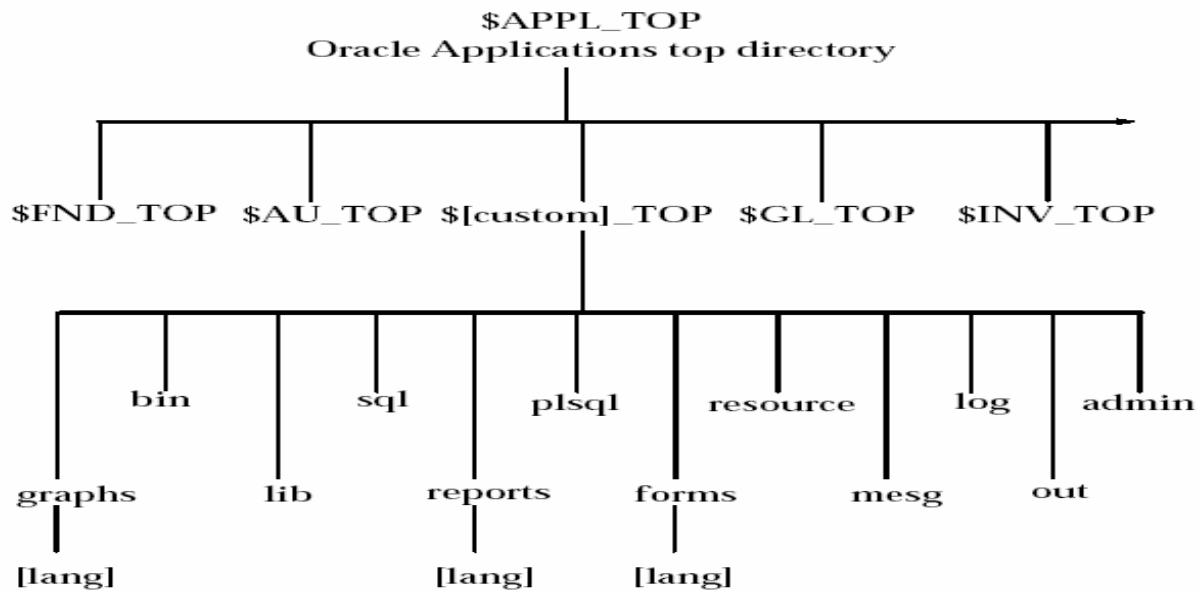
- An Oracle Applications Release 11i system utilizes components from many Oracle products.
- These product files are stored below a number of key top-level directories on the database and application server machines.
- File systems in detail, including the following:
 - Top-Level Directories
 - Product Directories
 - Environment Settings

Top-Level Directories:



- The **<dbname>DATA or DATA_TOP** directory is located on the database server and it contains the system tablespaces, redo log files, data tablespaces, index tablespaces, and database files.
- The **<dbname>DB directory** is located on the database server and it contains Oracle Home for the Oracle9i RDBMS.
- The **<dbname>APPL or APPL_TOP** directory contains the product directories and files for Oracle Applications.
- The **<dbname>ORA directory** contains the Oracle Homes for the Applications technology stack components.
- The **<dbname>COMN or COMN_TOP (or COMMON_TOP)** directory contains directories and files used across products.

Application Directory Structure



Data Directory

- This file system contains .dbf files of oracle db.
- Rapid installation installs system, data and index files in directories below several mount points (Can be specified during installation).

DB & ORA Directory

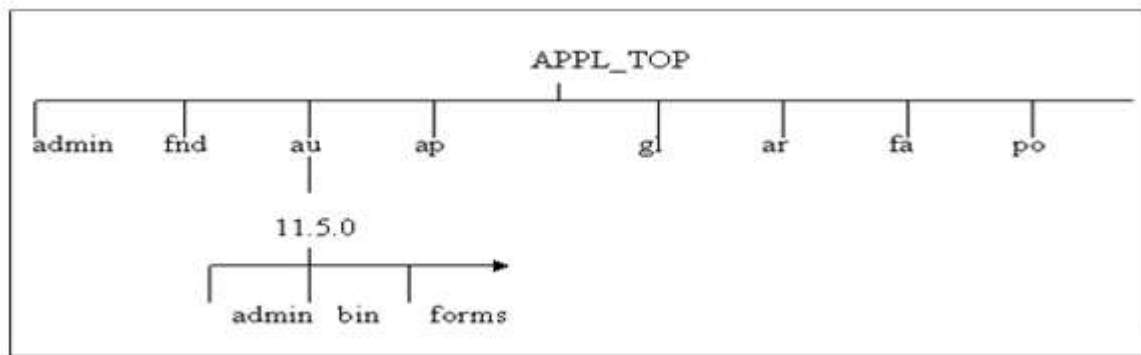
- Oracle application supports linking program using tool from 2nd / 3rd version of DB ORACLE_HOME is used for backward compatibility.

Application Directory

- All application files are stored here. It is also known as APPL_TOP. It contains:
- Core technology files.
- Product files & directories.
- .env application environment file (UNIX) & .cmdon (Windows).

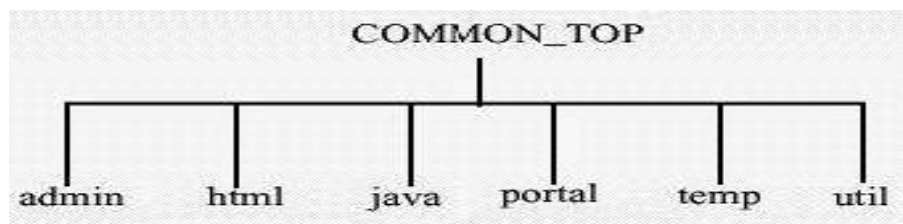
Product Directory

- Each product has its own directory under APPL_TOP.
- Sub Directories are named based on the product standard abbreviation. E.g. GL for general ledger.



COMN Directory

- COMN or COMMON_TOP directory contains files used by different oracle application products and used with third party products.



Admin

- It is default location for concurrent manager log and output directories.

Html

- OA_HTML environmental settings points to HTML Directory. Oracle application based sign-on screen & HTML files are installed here. Also it contains files like JSP, XML etc

Java

- JAVA_TOP points to java directory. This contains oracle application JAR files & ZIP Files.

Portal

- It has portal files. Portal is a webpage that provide access to post install task that is necessary for application

Temp

- Used for caching oracle products like reports.

Util

- This contains JDE, JRE & ZIP utility.

3. Explicate about oracle workflow analysis.

Oracle Workflow provides an infrastructure for the enterprise-wide communication of data related to defined business events, providing the capabilities needed to:

- Manage enterprise business processes that may span trading partners
- Support standard and personalized business rules
- Streamline and automate transaction flows • Manage exceptions without manual intervention
- Acting as a system integration hub, Oracle Workflow can apply business rules to control objects and route them between applications and systems.

Five Major Components of Workflow

- Oracle Workflow Builder
- Workflow Engine
- Workflow Definitions Loader
- Notification System
- Workflow Monitor

Oracle Workflow Builder

- The Workflow Builder is where the user: Creation views and modifies business process definitions graphically.
- Defines the activities and underlying components of business processes
- Graphically creates a process diagram by assembling the activities in the process window
- Workflow Builder allows the user to save a process diagram to a database or a flat file
- The Workflow Builder consists of:
 - Object Navigator (Tree structure)
 - Process Diagram.

Oracle Workflow Engine

- Embedded in the Oracle8/8i server
- Monitors Workflow statuses
- Coordinates the routing of activities for a process
- Contains all packages and procedures

- Used in Workflow
- Workflow process definition MUST be saved to the same database as the Workflow Engine.

Oracle Workflow Definitions Loader

- A utility program that allows the user to transfer process definition between a database and a flat file
- Runs on the server
- Integrated with Oracle Workflow Builder
- Users do not have to be on-line to work.

Workflow Notification System

- Receive and respond to notifications using e-mail or a Web browser
- With available Web functionality, all users with access to the internet can be included in the workflow process
- Access to Notification Workflow from Oracle Applications
- A list of allowed responses if a response is required

Notification Process Flow

- Workflow Engine Stops
- Notification system responds to trigger by sending note
- Notification system notifies Workflow Engine to continue process.

Workflow Monitor

- Notifications List
 - Displays all current notifications
 - Shows the decision makers in a process as well as the current owner of a process.
- Features
 - Graphically depicts the status of a workflow process
 - Displays detailed information about individual activities

4. Detail about fundamentals of SQL/PLSQL.

SQL:

- SQL (pronounced sequel) is the set-based, high-level declarative computer language with which all programs and users access data in an Oracle database.
- SQL provides an interface to a relational database such as Oracle Database. SQL unifies tasks such as the following in one consistent language:
 - Creating, replacing, altering, and dropping objects.
 - Inserting, updating, and deleting table rows.
 - Querying data.
 - Controlling access to the database and its objects.
 - Guaranteeing database consistency and integrity.

SQL Statements

- All operations performed on the information in an Oracle database are run using SQL statements.
- A SQL statement is a computer program or instruction that consists of identifiers, parameters, variables, names, data types. The types of statements are
 - Data Definition Language
 - Data Manipulation Language

Data Definition Language (DDL) Statements:

- Data definition language (DDL) statements used to Create, alter, and drop schema objects and other database structures, including the database itself and database users. Most DDL statements start with the keywords CREATE, ALTER, or DROP.
- Delete all the data in schema objects without removing the structure of these objects (TRUNCATE).
- Grant and revoke privileges and roles (GRANT, REVOKE).
- Turn auditing options on and off (AUDIT, NOAUDIT).
- Add a comment to the data dictionary (COMMENT).

Data Manipulation Language (DML) Statements:

- Data manipulation language (DML) statements query or manipulate data in existing schema objects.
- DML statements are the most frequently used SQL statements and enable to:
- Retrieve or fetch data from one or more tables or views (SELECT).

- Add new rows of data into a table or view (INSERT) by specifying a list of column values or using a subquery to select and manipulate existing data.
- Change column values in existing rows of a table or view (UPDATE).
- Update or insert rows conditionally into a table or view (MERGE).
- Remove rows from tables or views (DELETE).
- View the execution plan for a SQL statement (EXPLAIN PLAN). See "How Oracle Database Processes DML".
- Lock a table or view, temporarily limiting access by other users (LOCK TABLE).

PL/SQL

- PL/SQL stands for Procedural Language extension of SQL. PL/SQL is a combination of SQL along with the procedural features of programming languages.
- It was developed by Oracle Corporation in the early 90's to enhance the capabilities of SQL.

PL/SQL Engine:

- A PL/SQL language code can be stored in the client system (client-side) or in the database (server-side).
- The concepts like cursors, functions and stored procedures can be used in other database systems like Sybase, Microsoft SQL server etc, with some change in SQL syntax.

PL/SQL Block consists of three sections:

- The Declaration section (optional).
- The Execution section (mandatory).
- The Exception Handling (or Error) section (optional).

Declare Variable declaration Begin Program execution Exception Exception handling End

PL\SQL block

Declaration Section:

- The Declaration section of a PL/SQL Block starts with the reserved keyword DECLARE.

- It is optional and is used to declare any placeholders like variables, constants, records and cursors used to manipulate data in execution section.

Execution Section:

- The Execution section of a PL/SQL Block starts with the reserved keyword BEGIN and ends with END.
- This is a mandatory section and is the section where the program logic is written to perform any task

Exception Section:

- The Exception section of a PL/SQL Block starts with the reserved keyword EXCEPTION.
- It is optional, any errors in the program can be handled in this section, so that the PL/SQL Blocks terminates gracefully.
- If the PL/SQL Block contains exceptions that cannot be handled, the Block terminates with errors.

PL/SQL Subprograms

- A PL/SQL subprogram is a named PL/SQL block that permits the caller to supply parameters that can be input only, output only, or input and output values.
- A subprogram is either a procedure or a function.
- Procedures and functions are identical except that functions always return a single value to the caller, whereas procedures do not. The term procedure in this chapter means procedure or function.

Advantages of PL/SQL Subprograms

- Improved performance
- Memory allocation
- Improved productivity
- Integrity
- Security with definer's rights procedures.

5. Describe in detail about creating forms and oracle reports.

FORMS:

- Oracle Forms provides a mechanism to store procedures (called Program Units) written in the PL/SQL language within a form. By storing a Program Unit within a form, many blocks in the form can take advantage of the same procedure which reduces code duplication and improves the maintainability of the code.

- Procedures written in PL/SQL may also be stored within the Oracle Database itself as an object in the schema. Such Stored Program Units (also called Stored Procedures) are ideal for situations where highly standardized business rules or applications logic must be implemented across many forms or applications.
- In addition, procedures that require several queries be made to the database may be best implemented in the database as this reduces the network traffic and can significantly improve performance.
- Oracle Forms can make calls to both internal Program Units as well as to Stored Program Units. This flexibility allows application designers extensive control over the execution of applications and facilitates performance tuning.

ORACLE FORMS DESIGN

- In Oracle Forms, a form (or data entry form) acts like a window into the database schema. An individual form focuses the attention of the user to one or a few of the tables at a time.
- A form can give prompts so the user knows what kind of input is expected and how data is to be entered and manipulated.
- We can program forms with additional business logic that can help detect bad data or mistakes, make automatic calculations, and suggest appropriate data to be entered and many other helpful techniques that can ensure high quality data can be easily entered into the database.
- By default, every form in Oracle Forms has the capability to query existing data in a table, modify existing data and add new data (records) to the table. A form is built up using one or more data blocks that correspond to tables in the database.
- Fields within the data block correspond to columns in the database table. A data block is similar to a Data Window object.

EXAMPLE

- The Example form (in fig) has two data blocks, one for the EMPLOYEE table, and one for the DEPENDENT table. These data blocks are arranged in a Master/Detail setup where a single Employee record (the master) is associated with one or more Dependents records (the details). In this database, Dependents are the spouse, sons and daughters of the employee.

- By default, forms also give a button bar and a menu. These can be used to scroll through the records in a block, navigate between blocks, set up and perform queries against the tables, insert, and update and delete records, clear the form and exit the form. Finally, at the bottom of each form is a status bar that displays any relevant prompts or error messages and an indication of the records in the current data block.

Oracle9iAS Forms Services - Netscape

File Edit View Go Communicator Help

Bookmarks Location: g_messages=NO&array=YES&query_only=NO&quiet=NO&RENDER=YES

Menu Bar ORACLE

Action Edit Query Block Record Field Help Window

Button Bar

WINDOW1

Employee

First Name: JOHN

Last Name: SMITH

SSN: 123456789

Address: 731 FONDREN, HOUSTON, TX

Dependents

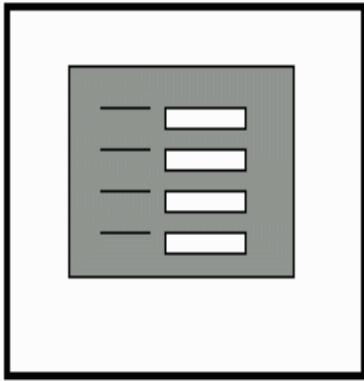
Dependent Name	Birth Date	Relationship
ALICE	31-DEC-1978	DAUGHTER
ELIZABETH	05-MAY-1957	SPOUSE
MICHAEL	01-JAN-1978	SON

Data Blocks

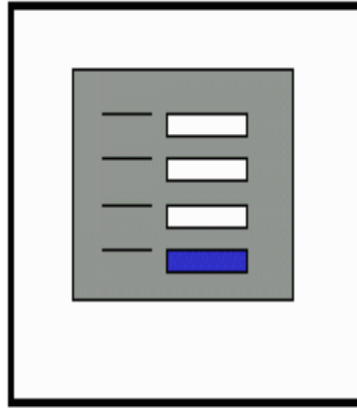
Status Bar

Record: 4/4

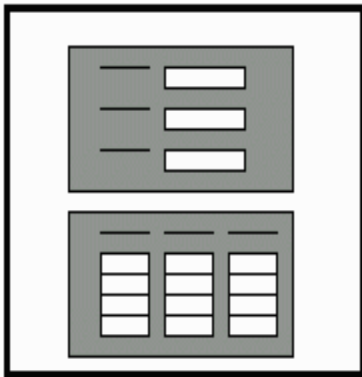
There are four main types of forms that can be designed.



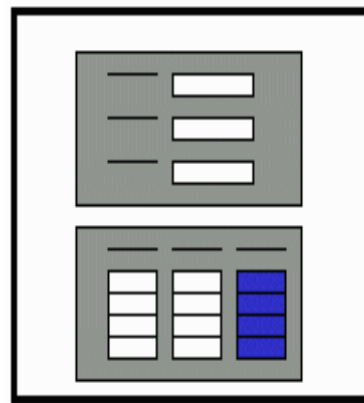
Single Block Form. This form contains a single data block corresponding to a single database table.



Single block Form with lookup field. This form contains a single data block corresponding to a single database table with the addition of one or more fields that display data from some other tables. Such data is “looked up” when the form runs.



Master/Detail Form. This form contains two data blocks that are arranged in a master/detail (one to many) relationship.



Master/Detail Form with lookup fields. This form is similar to the Master/Detail but has the additional lookup fields in the master and/or detail blocks.

Connect Oracle Forms 11g to database 11g

1. First you need to copy the file tnsnames.ora from database folder to config folder from Oracle Middleware.
2. Open the Oracle Form Builder, go to File Menu and press Connect or just execute the command Ctrl+J.
3. On the last step, you need to enter the user name, password and the database name.
4. Change Module Name in Oracle Forms 11g

Right-click on Form Module name, then choose Property Palette. On Property Palette just click on the name and write your module name.

5. Open the Oracle Form Builder, go to File Menu and press Connect or just execute the command Ctrl+J.

6. Create New Data Block Wizard in Oracle Forms 11g

First you must click on Data Blocks and then press the button Create. An alert with two options will appear, and here you choose: Use the Data Block Wizard.

7. Welcome to the Data Block Wizard! - Click Next to begin creating your data block.

8. Select the type of Data Block: Table/View or Stored Procedure.

9. Create New Data Block Wizard in Oracle Forms 11g

Press Browse button and will appear a small window with more choices: Tables, Views, Current user, Synonyms, Other users.

10. After you choose the table name, Oracle Forms will show all Available Columns from the table.

Here you can bring all columns from the table in your form or you can select just a few columns.

11. After you moved the items press Next.

12. Create New Data Block Wizard in Oracle Forms 11g

Enter a name for your Data Block.

13. You have finished describing your data block. Now you will have two options:

- Continue with: Call the Layout Wizard
- Stop: Just create the data block (is recommended when you use multiple data block on the same canvas)

14. Create Wizard Layout Oracle Forms 11g: Display in a frame on a canvas

15. Here you have to choose the canvas name (can be a new one or an existing canvas) and the type of canvas.

Types of canvas:

- Content
- Stacked
- Vertical Toolbar
- Horizontal Toolbar
- Tab

16. Create Wizard Layout Oracle Forms 11g

Here you select and move item that you want to be displayed in the frame.

For each each item displayed you can choose the item type from the list.

17. Enter a prompt, width and the height for each item.

18. Create Wizard Layout Oracle Forms 11g

Select the Layout Style for your frame

19. On this step you can choose the Frame Title, you can insert how many records to be displayed in the field Records Displayed, also Distance Between Records and Display Scrollbar.

20. Finish the Wizard Layout and display the new frame in the Layout Editor

You Finish the Wizard Layout.

21. Display the new frame in the Layout Editor.

REPORTS:

Oracle Reports is the enterprise reporting component of Oracle's BI offering. It enables an organization to convert data into information, and publish the information secure and reliable. The Oracle Reports Builder allows the developer to create sophisticated reports in a variety of layouts and contains many customization features.

- Oracle Reports provides an access to multiple data sources. E.g: manufacturing organization
- It supports SQL, PL/SQL, XML, JDBC, Oracle OLAP, and text files. JDBC data source can be used to access data from different JDBC-compliant data sources, and JDBC-ODBC Bridge can be used to connect to any data that can be exposed as an ODBC data source.
- With Oracle Reports, the source of data is transparent, because data coming from any data source is available to you in exactly the same representation.

Features of Oracle Reports

- The most flexible enterprise-reporting tool in the market today
- Reliable
- Integrated
- Extensible

The Main section of Object navigator is Reports

Reports which include:

1. **Data Model** – Contains information about queries used for a report.
2. **Web Source** – A web page or web service that returns a data set.

3. **Paper Layout** – Contains information about how a paper or screen a report is formatted including headers, footers, margins, fonts, etc.
4. **Paper Parameter Form** – Contains information about the initial screen that is displayed when a form first runs.
5. **Report Triggers** – PL/SQL code that can be executed before, during or after a report has been executed.
6. **Program Units**
7. **Attached Libraries**

Creating reports follows a 4 step process:

1. **Define the Data Model** – This step specifies which queries should be run on the database including how multiple queries are related and how they are grouped. This step must be done by hand. Queries that have been created elsewhere can be imported into Oracle Reports.
2. **Define the Layout** – This step specifies the layout of the report including the overall orientation of query results and the suppression of repeating groups. There are a number of default report layouts that can automatically be applied to a data model.
3. **Create and/or Customize the Parameter Form** – If some user input is required in order to run the report, then a parameter form must be customized. All reports have a default parameter form.
4. (optional) **Create any triggers or program units that will be executed with the report.**

How to generate Report?

1. Connect Oracle Reports 11g to database 11g
Open the Oracle Reports Builder, go to File Menu and press Connect or execute the command Ctrl+J.
2. On the second step, you need to enter the user name, password and the database name.
Create Report Wizard in Oracle Reports 11g
First you must click on Reports and the press the button Create. An alert "Create a new report" with two option will appear, and here you choose: Use the Report Wizard.
3. Welcome to the Report Wizard! - Click Next to begin creating your report.
4. Choose The Type of Layout and Style Reports in Oracle 11g
Here you must choose the type of layout you would like to generate: Create both Web and Payper Layout, Create Web Layout only, Create paper Layout only.

5. On the next step you have to choose the report style and give a title name for your report. The report style options are: Tabular, Group Left, Group Above, Matrix, Matrix with Group.
6. Choose a Data Source and Select Sql Query Builder for Reports in Oracle 11g
On this step you have to choose a Data Source for your report. The Data Source can be from JDBC Query, SQL Query, Text Query, XML Query.
7. You can do write the query manually, you can use Query Builder or Import Builder.
8. Move Available Fields in Displayed Fields in R reports Oracle 11g
Select the fields that you would like to display in your report. You can move one item or all items to target from Available Fields in Displayed Fields.
9. After you moved the items press Next.
10. Select Available Fields in Reports Wizard to calculate totals
Select the fields that you would like to calculate totals in your report. Also you can do choose to calculate: sum, count, minimum, maximum, average.
11. Here you can modify the labels of the fields and also the width for each field.
12. Choose a Template for your Report Wizard in Oracle Reports 11g
Select the template for your report. You have multiple options: Predefined Template, Template File or No Template.
13. You have finished describing your report!
14. Run Module Report Editor Paper Design Oracle Reports 11g
Select the template for your report. You have multiple options: Predefined Template, Template File or No Template.

6. Explain about oracle Electronic Data Interchange (EDI) and its functions.

- EDI (Electronic Data Interchange) is the electronic exchange of business documents between the computer systems of business partners, with a standard format over a communication network.
- It can also be called as paperless exchange.
- EDI, electronic funds transfer (EFT), electronic mail and fax are increasingly being used to schedule operations, streamline order fulfillment, and optimize cash flow. The National Institute of Standards and Technology in a 1996 publication defines Electronic Data Interchange as "the computer-to-computer interchange of strictly formatted messages that represent documents other than monetary instruments."

- EDI can be transmitted using any methodology agreed to by the sender and recipient. This includes a variety of technologies, including modem (asynchronous, and bisynchronous), FTP, Email, HTTP etc.

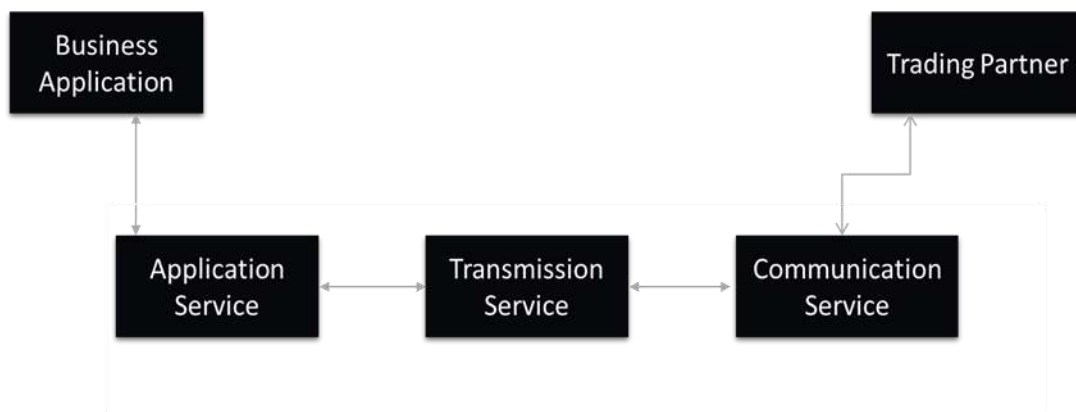
EDI Standard Formats

- The business documents that are been exchanged between business partners need to be in a standard format. ANSI X12 (American National Standards Institute) or EDIFACT (Electronic Data Interchange For administration, Commerce, and Transport) are two standards, which supply a common language for formatting the information content that is been exchanged.

Components

Three major Components in the EDI Process:

1. Application service
2. Translation service
3. Communication service



Application Services:-

- It provides the link between application and EDI. It allows you to send documents from an EDI system.
- The set of callable routine is used to transfer document from the business application into EDI document destination can be either intra-company or to the external companies.

Translation service:-

- Converts the outgoing documents from an internal format file to an agreed external format. Translates internal document from external format to EDI internal format file.

Communication service:-

- The communication service sends and receives transmission files to and from the trading partners either directly or by using party service called a valued added network (VAN).

File Types

- **Internal format file (IFF):-**It contains single document for single trading partner.
- **External format file (EFF):-**It contains same data as the internal format file translated into the appropriate standard document format.
- **Transmission file:-** it contains one or more document for the same trading partner. Documents of same format are packed into functional groups. The functional groups going to one trading partner are packaged into an interchanged set. An interchanged set contains one or more functional groups of documents with the same sender and receiver.

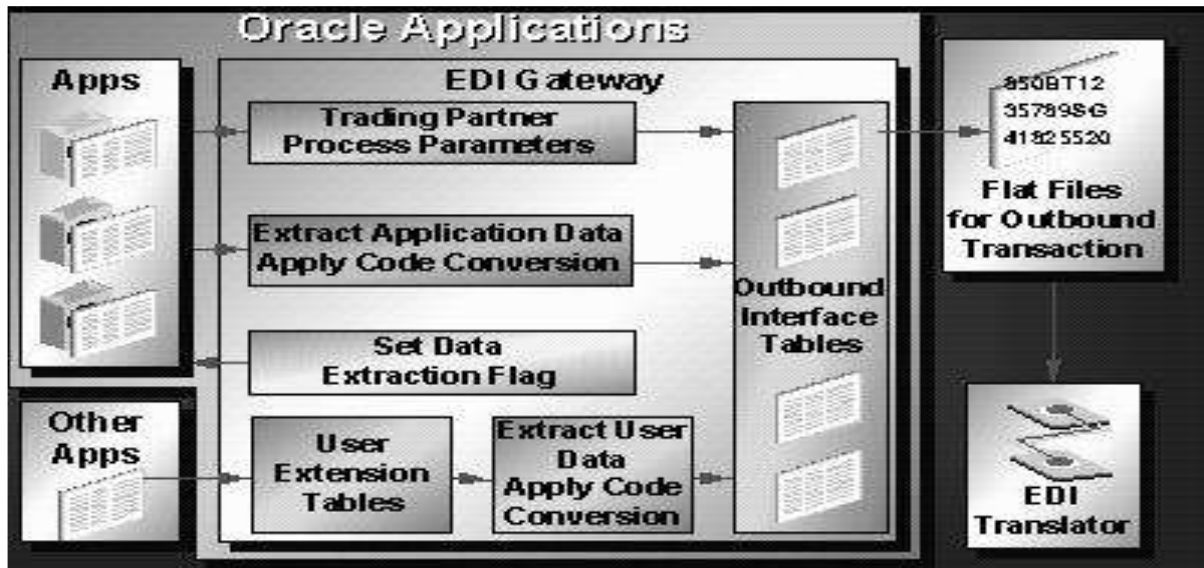
Advantages

- One Time Data Entry
- Reduced Errors
- Online Data Storage
- Faster management Reporting
- Automatic Data Reconciliation

FUNCTIONS OF EDI

- Define Trading Partner Relationships
- Define Code Conversions
- Customize Flat File Formats
- Use Standard Report Submission
- Initiate Extract Process
- Initiate Import Process
- View Extract/Import Process Status

A typical flow can be understood as:



Step 1. Define Data File Directories

- To use EDI Gateway, you must first create directories where data files, for both inbound and outbound transactions, are stored.

Step 2. Define Trading Partner Data

- First, define a trading partner group and associate all trading partner location (address) entities to this group. Define basic information about the trading partner and associate that to a specific trading partner's location.

Step 3. Define Code Conversions

- The Oracle EDI Gateway code conversion function provides a method by which trading partner and standard codes can be converted to Oracle Application codes.
- Define code conversion categories to specify a subset of codes within the code conversion table.
- For each data element in a transaction that requires code conversion by EDI Gateway, assign one code conversion category.
- Enter the actual code conversion values from the internal codes to external codes into the code conversion table.

Step 4. Customize Data File Formats

- Oracle delivers a standard predefined data file format containing all the data that may be required to support a particular EDI transaction.

Step 5. Initiate EDI Transactions

- Once data file directories, trading partner information, code conversions, and optional customizations of data files have been performed, use Oracle Applications Standard Request Submission to run extract programs for outbound transactions and import programs for inbound transactions.

7. Elucidate in detail about oracle data and oracle database.

Oracle database

Database Management System (DBMS)

- A database management system (DBMS) is software that controls the storage, organization, and retrieval of data.
- A database application is a software program that interacts with a database to access and manipulate data.
- A relational database stores data in a set of simple relations. A relation is a set of tuples. A tuple is an unordered set of attribute values.
- A table is a two-dimensional representation of a relation in the form of rows (tuples) and columns (attributes).

History of Oracle Database

- Founding of Oracle: In 1977, Larry Ellison, Bob Miner, and Ed Oates started the consultancy Software Development Laboratories, which became Relational Software, Inc. (RSI). In 1983, RSI became Oracle Systems Corporation and then later Oracle Corporation.
- First commercially available RDBMS: In 1979, RSI introduced Oracle V2 (Version 2) as the first commercially available SQL-based RDBMS.
- Portable version of Oracle Database: Oracle Version 3, released in 1983, was the first relational database enabling the database to be ported to multiple platforms.
- Enhancements to concurrency control, data distribution, and scalability
- Version 4 introduced multiversion read consistency. Version 5, released in 1985, supported client/server computing and distributed database systems. Version 6 brought enhancements to disk I/O, row locking, scalability, and backup and recovery.

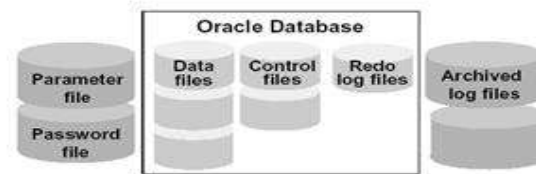
ORACLE DATABASES

- Oracle Database (commonly referred to as Oracle RDBMS or simply as Oracle) is an object-relational database management system produced and marketed by Oracle Corporation.
- Larry Ellison and two friends and former co-workers, Bob Miner and Ed Oates, started a consultancy called Software Development Laboratories (SDL) in 1977. SDL developed the original version of the Oracle software

Oracle Database

An Oracle database:

- Is a collection of data that is treated as a unit
- Consists of three file types



Physical Structures

- **Datafiles (*.dbf)**

The datafiles contain all the database data. The data of logical database structures, such as tables and indexes, is physically stored in the datafiles allocated for a database

- **Control Files (*.ctl)**

A control file contains entries that specify the physical structure of the database such as Database name and the Names and locations of datafiles and redo log files

- **Redo Log Files (*.log)**

The primary function of the redo log is to record all changes made to data. If a failure prevents modified data from being permanently written to the datafiles, then the changes can be obtained from the redo log, so work is never lost

- **Archive Log Files (*.log)**

Oracle automatically archives log files when the database is in ARCHIVELOG mode. This prevents oracle from overwriting the redo log files before they have been safely archived to another location.

- **Parameter Files (initSID.ora)**

Parameter files contain a list of configuration parameters for that instance and database.

- **Alert and Trace Log Files (*.trc)**

Each server and background process can write to an associated trace file. When an internal error is detected by a process, it dumps information about the error to its trace file.

Logical Structures

- **Tablespace**

A database is divided into logical storage units called tablespaces, which group related logical structures together. One or more datafiles are explicitly created for each tablespace to physically store the data of all logical structures in a tablespace.

- **Data Blocks**

At the finest level of granularity, Oracle database data is stored in data blocks. One data block corresponds to a specific number of bytes of physical database space on disk.

- **Extents**

An extent is a specific number of contiguous data blocks, obtained in a single allocation, used to store a specific type of information.

- **Segments**

A segment is a set of extents allocated for a certain logical structure. The different types of segments are :

1. Data segment – stores table data
2. Index segment – stores index data
3. Temporary segment – temporary space used during SQL execution
4. Rollback Segment – stores undo information

- **Schema**

A schema is a collection of database objects. A schema is owned by a database user and has the same name as that user. Schema objects are the logical structures that directly refer to the database's data. Schema objects include structures like tables, views, and indexes

Oracle Instance

- Logical memory structures that are collectively called an Oracle instance
- Physical file structures that are collectively called a database .It is the actual execution of DBMS software that manages data in the databases tablespace. Two Main Components:

SGA (System Global Area)

- A group of shared memory structures that contain data and control information for one Oracle database instance.

- The data in the instance's SGA is shared among the multiple concurrent users.
- Allocated when you start the database instance.
- De-allocated when the instance is shutdown.

PGA (Program Global Area)

- Each server process has a PGA allocated that is a private area for each server
- Work area for each application.

8. Differentiate DW vs OLTP.

Characteristics	Data Warehouse	OLTP
Source of data	Consolidation data; OLAP data comes from the various OLTP Databases	Operational data; OLTPs are the original source of the data.
Purpose of data	To help with planning, problem solving, and decision support	To control and run fundamental business tasks
What the data	Multi-dimensional views of various kinds of business activities	Reveals a snapshot of ongoing business processes
Inserts and Updates	Periodic long-running batch jobs refresh the data	Short and fast inserts and updates initiated by end users
Queries	Often complex queries involving aggregations	Relatively standardized and simple queries Returning relatively few records
Processing Speed	Depends on the amount of data involved; batch data refreshes and complex queries may take many hours; query speed can be improved by creating indexes	Typically very fast
Space Requirements	Larger due to the existence of aggregation structures and history data; requires more indexes than OLTP	Can be relatively small if historical data is archived
Database Design	Typically de-normalized with fewer tables; use of star and/or snowflake schemas	Highly normalized with many tables
Backup and Recovery	Instead of regular backups, some environments may consider simply reloading the OLTP data as a recovery method	Backup religiously; operational data is critical to run the business, data loss is likely to entail significant monetary loss and legal liability
Workload	Designed to accommodate adhoc queries. We may not know the workload of our data warehouse in	Supports only predefined operations. Our applications might be specifically tuned or designed to support only these

	advance, so it should be optimized to perform well for a wide variety of possible queries	operations.
Data Modifications	Updated on a regular basis by the ETL process using bulk data modification techniques. End user of a data warehouse do not directly update the database	Subjected to individual DML statements routinely issued by end users. The OLTP database is always up to date and reflects the current state of each business transaction.
Scheme design	Uses denormalized or partially denormalized schemes (such as star scheme) to optimize query performance.	Uses fully normalized schemes to optimize DML performance and to guarantee data consistency

9. Discuss in detail about DW Connectors.

Teradata OLAP Connector

- Teradata OLAP Connector is the one tool available that makes the direct connection between the Teradata system and Excel applications.

Overview

- Teradata OLAP Connector provides self-service analytics for Excel, Tableau, and arcplan users by enabling them to access the corporate data stored in the Teradata Data Warehouse.
- This eliminates the inconsistencies and security concerns associated with the traditional approach of extracting and storing corporate data locally.

Challenges

Desktop-based BI solutions allow business users to extract data from the data warehouse for desktop analysis but this approach is challenging:

- Creates spread marts all over the organization
- Static view of the data of when it was extracted which leads to inconsistencies with the reporting
- Limited data leads to limited analysis
- Causes security challenges for leaving sensitive data on the users' desktops
- Data extractions scripts are difficult to maintain

ORACLE BIG DATA CONNECTORS

- Oracle Big Data Connectors is a software suite that integrates processing in Hadoop with operations in a data warehouse.

- Designed to leverage the latest features of Apache Hadoop, Big Data Connectors connect Hadoop clusters with database infrastructure to harness massive volumes of structured and unstructured data for critical business insights.
- Big Data Connectors greatly simplify development and are optimized for efficient connectivity and high-performance between Oracle Big Data Appliance and Oracle Exadata. Oracle Big Data Connectors 3.0 delivers a rich set of new features, increased connectivity, enhanced performance, and security for Big Data applications.

Oracle Big Data Connectors

- Large volumes of data are increasingly collected and processed in Hadoop, while enterprise IT systems are centered on relational data warehouses.
- Oracle Big Data Connectors bridges data processing in Hadoop with Oracle Database, providing the crucial ability to unify data across these systems.
- Combining pre-processing of large data volumes of raw and unstructured data in Hadoop with the advanced analytics, complex data management, and real-time query capabilities of Oracle Database, Oracle Big Data Connectors deliver features that support information discovery, deep analytics and fast integration of all data in the enterprise.
- The components of this software suite are:
 - Oracle SQL Connector for Hadoop Distributed File System
 - Oracle Loader for Hadoop
 - Oracle Data Integrator Application Adapter for Hadoop
 - Oracle R Advanced Analytics for Hadoop
 - Oracle XQuery for Hadoop
 - Oracle Big Data Connectors work with Oracle's engineered systems - Oracle Big Data Appliance and Oracle Exadata - as well as with supported Hadoop distributions and database versions on non-engineered systems.

KEY FEATURES

- Tight integration with Oracle Database
- Leverage Hadoop compute resources for data in HDFS
- Enable Oracle SQL to access and load Hadoop data
- Fast and very efficient load from Hadoop into Oracle Database

- Partition pruning of Hive tables during load and query
- Graphical user interfaces of Oracle Data Integrator drive data transformation workflows on.

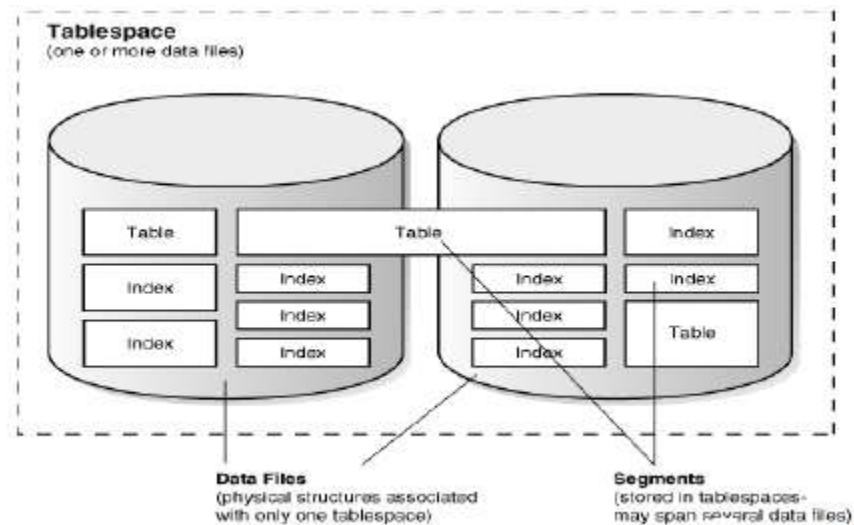
10. Explain in detail about Data file structure.

Overview of Data Files

- At the operating system level, Oracle Database stores database data in data files.
- Every database must have at least one data file.

Use of Data Files

- Oracle Database allocates space for user data in tablespaces, the data for a nonpartitioned table is stored in a single segment, which is turn is stored in one tablespace..
- Oracle Database physically stores tablespace data in data files
- Each tablespace consists of one or more data files, which conform to the operating system in which Oracle Database is running.
- The data for a database is collectively stored in the data files located in each tablespace of the database.
- A segment can span one or more data files, but it cannot span multiple tablespaces.
- A database must have the SYSTEM and SYSAUX tablespaces. Oracle Database automatically allocates the first data files of any database for the SYSTEM tablespace during database creation.



Relationship between table spaces, data files, and segments.

Permanent and Temporary Data Files

- A permanent tablespace contains persistent schema objects. Objects in permanent tablespaces are stored in data files.
- A temporary tablespace contains schema objects only for the duration of a session.

Online and Offline Data Files

- Every data file is either online (available) or offline (unavailable). You can alter the availability of individual data files or temp files by taking them offline or bringing them online.
- Offline data files cannot be accessed until they are brought back online.
- Administrators may take data files offline for many reasons, including performing offline backups, renaming a data file, or block corruption.
- The database takes a data file offline automatically if the database cannot write to it.

Data File Structure

- Oracle Database creates a data file for a tablespace by allocating the specified amount of disk space plus the overhead for the data file header.
- The operating system under which Oracle Database runs is responsible for clearing old information and authorizations from a file before allocating it to the database.
- The data file header contains metadata about the data file such as its size and checkpoint SCN. Each header contains an absolute file number and a relative file number. The absolute file number uniquely identifies the data file within the database.
- The relative file number uniquely identifies a data file within a tablespace. When Oracle Database first creates a data file, the allocated disk space is formatted but contains no user data.
- However, the database reserves the space to hold the data for future segments of the associated tablespace. As the data grows in a tablespace, Oracle Database uses the free space in the data files to allocate extents for the segment. Over time, updates and deletions of objects within a tablespace can create pockets of empty space that individually are not large enough to be reused for new data.
- This type of empty space is referred to as fragmented free space.



Space in a Data File